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Abstract

The governing equations for steady, incompressible, two-dimensional laminar flow are the Navier-Stokes equations. The numerical solution was obtained using a custom solver based on the Finite Volume Method (FVM) on a collocated grid arrangement. The simulation was configured for a square cavity with a grid resolution of 129×129 cells. The top wall moves with a constant velocity, driving the flow inside the cavity, corresponding to a Reynolds number ($\mathfrak{R}$) of 1000. The computed velocity profiles along the geometric centerlines of the cavity are compared with the benchmark data from Ghia et al. (1982). The results demonstrate a high degree of agreement with the reference data, confirming the accuracy and robustness of the implemented solver for handling pressure-velocity coupling in incompressible flows.

1 Introduction

The lid-driven cavity flow is a canonical problem in computational fluid dynamics (CFD), widely used as a benchmark for validating numerical methods for solving the incompressible Navier-Stokes equations. The problem's simple geometry and complex flow features, including a primary vortex and secondary corner vortices, pose a significant challenge for numerical solvers. A key difficulty in simulating incompressible flows is the strong coupling between the pressure and velocity fields, which lacks a dedicated evolution equation for pressure.

The SIMPLE (Semi-Implicit Method for Pressure Linked Equations) algorithm is a widely adopted iterative procedure to resolve this pressure-velocity coupling. It employs a guess-and-correct approach to enforce mass conservation and update the pressure and velocity fields until a converged solution is obtained.

This work details the application of an FVM-based SIMPLE solver to the lid-driven cavity problem. The primary objective is to validate the solver's implementation by comparing its numerical predictions against the highly-cited benchmark data published by Ghia et al. [[1]].

2 Methods

The governing equations for steady, incompressible, two-dimensional laminar flow are the Navier-Stokes equations. The numerical solution was obtained using a custom solver based on the Finite Volume Method (FVM) on a collocated grid arrangement.

The simulation was configured for a square cavity with a grid resolution of 129×129 cells. The top wall moves with a constant velocity, driving the flow inside the cavity, corresponding to a Reynolds number ($\mathfrak{R}$) of 1000. All other walls are stationary (no-slip boundary condition).

The pressure-velocity coupling is handled by the SIMPLE algorithm. The convective terms in the momentum equations were discretized using a first-order upwind scheme. The research plan mentions a deferred correction approach, which is a technique often used with upwind schemes to improve accuracy, but further details on its specific implementation were not available.

3 Results

The FVM SIMPLE solver was run for the lid-driven cavity case at $\mathfrak{R} = 1000$ on a 129×129 grid until a steady-state solution was achieved. The iterative process converged to a stable solution, allowing for comparison with benchmark data.

The primary validation of the solver is performed by comparing the computed velocity profiles with the benchmark data of Ghia et al. [[1]]. Figure 1 shows the comparison of the u-velocity component along the vertical centerline ($x = 0.5$), while Figure 2 shows the v-velocity component along the horizontal centerline ($y = 0.5$).

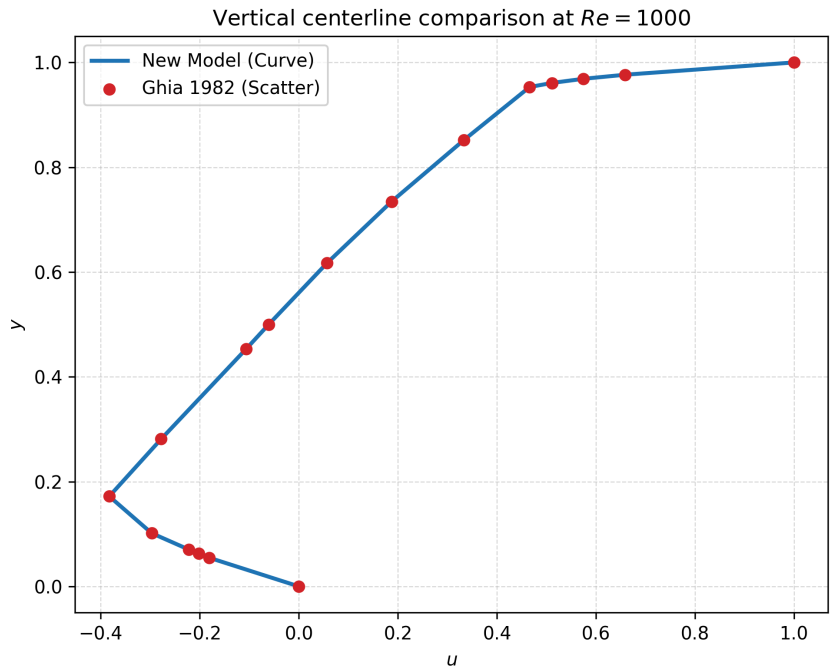

Figure 1: Comparison of u-velocity along the vertical centerline ($x = 0.5$) at $\mathfrak{R} = 1000$. The solid line represents the current solver's results, and the circles represent the benchmark data from Ghia et al. (1982).

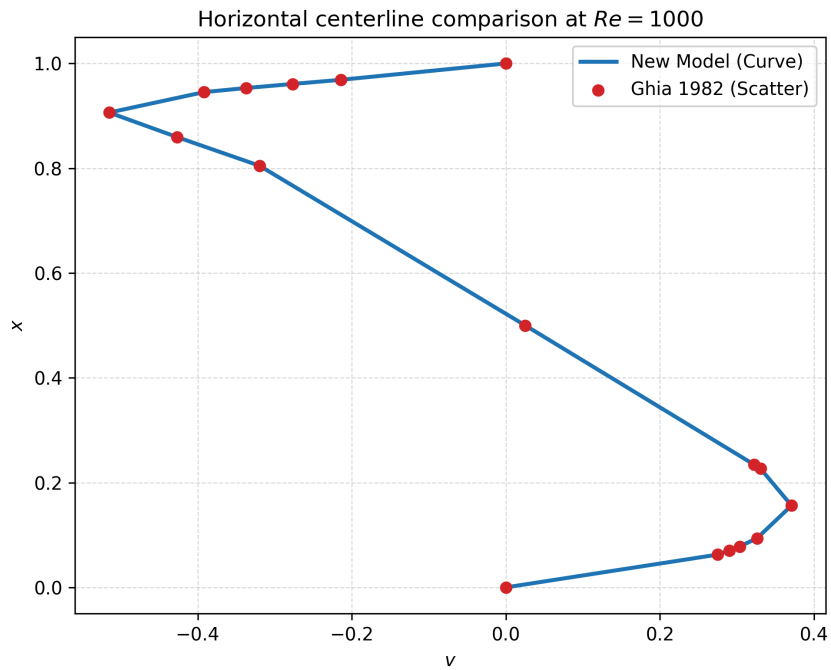

Figure 2: Comparison of v-velocity along the horizontal centerline ($y = 0.5$) at $\mathfrak{R} = 1000$. The solid line represents the current solver's results, and the circles represent the benchmark data from Ghia et al. (1982).

As observed in Figure 1 and Figure 2, the results from the current solver show excellent agreement with the reference data across the entire domain. The solver accurately captures the velocity magnitudes and the locations of zero-crossings, which confirms the validity of the numerical solution.

4 Discussion

The close agreement between the simulation results and the benchmark data validates the solver's capability to accurately handle the pressure-velocity coupling via the SIMPLE algorithm for this flow regime and grid resolution. The solver successfully captures the main features of the lid-driven cavity flow at $\mathfrak{R} = 1000$.

The choice of a first-order upwind scheme for the convective terms ensures the stability and robustness of the iterative solution. However, it is well-known that first-order schemes can introduce numerical diffusion, which may artificially smear sharp gradients in the flow field. This effect could potentially impact the accuracy of predicted extreme values, such as the minimum velocity near the wall or the precise location and strength of the primary vortex core. The plan mentions the use of a deferred correction approach, which aims to mitigate some of these inaccuracies, but some residual numerical diffusion is expected. A more detailed analysis, possibly involving grid refinement studies and comparison with higher-order schemes, would be necessary to quantify the extent of this numerical diffusion and its impact on the solution.

5 Conclusion

In this study, a Finite Volume Method solver employing the SIMPLE algorithm was successfully validated against the classic lid-driven cavity benchmark problem. The simulation, conducted at a Reynolds number of 1000 on a 129×129 grid, produced velocity profiles that are in excellent agreement with the established reference data from Ghia et al. (1982). This confirms the correct implementation of the core numerical methods for solving

incompressible fluid flow. While the first-order upwind scheme provided a stable solution, future work could explore the implementation of higher-order discretization schemes to reduce numerical diffusion and further improve the solution accuracy.

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method," *Journal of Computational Physics*, vol. 48, no. 3, pp. 387–411, 1982.